Day 03

Choosing the right model

Which model should we use, what will it cost, and how do we prove it's the right choice? The first question every client asks.

Closed vs open-weight | Licences | Reasoning vs fast | Cost per 1M tokens | Latency

Context size | Benchmarks lie | OpenRouter · Ollama · Hugging Face

who holds the weights?

Closed vs open-weight models

The difference is **access and control**, not intelligence. Both can summarise, code, do RAG, extract data and call tools.

Closed

Your app ⇄ Provider API ⇄ Their GPUs

- You send a request, they run the model
- You never get the weights
- Examples: **GPT, Claude, Gemini**

Open-weight

Your app ⇄ Ollama / vLLM ⇄ Your GPUs

- Download the weights, run them anywhere
- Or use a host that runs them for you
- Examples: **Llama, Qwen, DeepSeek, Mistral, gpt-oss**

careful: open-weight ≠ open-source. The weights are released, but training data and code usually aren't.

two separate questions

Which model × where it runs

	Hosted API (someone else's servers)	Self-hosted (your servers)
Closed	**GPT, Claude, Gemini via their own APIs** the normal way to use them	**Not possible** you never get the weights
Open-weight	**Open model on a host, e.g. OpenRouter** open weights, but no GPUs to manage	**Download weights, run on Ollama / vLLM** full control, full responsibility

AI startup, 2 weeks, no GPUs
→ closed model via API

Bank / hospital, data stays inside
→ open-weight, self-hosted

Millions of simple jobs
→ test both, do the math

trade-offs

Benefits and limitations

Closed

+ Ship fast: no GPUs, no model serving
+ Strong at hard reasoning, coding, tools
+ Scale up without buying hardware
+ Swap models easily to compare

– Pay for every token, forever
– Can't change the weights
– Provider risk: price changes, deprecations, outages
– Data leaves your network: check terms

Open-weight

+ Run anywhere: laptop, private cloud, on-prem
+ Fine-tune, quantise (licence permitting)
+ Prompts can stay inside your network
+ Can be cheaper at high, steady volume

– Big models need big, expensive GPUs
– You run ops: batching, scaling, monitoring
– Some are weaker at hard reasoning / tools
– Outages and regressions are your problem

remember: "the weights are free" ≠ "inference is free". GPUs, ops and engineering time cost money too.

can I actually use it?

Licences: permissive vs custom

Licence	Example (checked 2026-10-11)	What it means for you
Apache 2.0	gpt-oss-120b, Qwen3-8B, Mistral Small 4, Ministral 3	Use commercially, modify, redistribute. Keep the licence and notices. Includes a patent grant.
MIT	DeepSeek-R1	Very permissive. DeepSeek explicitly allows commercial use and distillation.
Custom	Llama 4 Community License	Over **700M monthly users**? Ask Meta first. Show **"Built with Llama"**. Fine-tuned models must have names starting with **"Llama"**.
Non-commercial	e.g. CC BY-NC 4.0 (Mistral's Voxtral TTS)	Fine for learning and research. **Not for a paid product.**

Check per model version, not per family: a DeepSeek-R1 distill built on Llama inherits Llama's licence.

Sources: HF model cards (openai/gpt-oss-120b, Qwen/Qwen3-8B, deepseek-ai/DeepSeek-R1) · docs.mistral.ai/getting-started/models · dev.meta.ai/llama/llama4/license

think first, or answer now?

Reasoning vs fast / small models

Fast / small

prompt → answer

Good for: classification, extraction, routing, simple summaries, JSON formatting

\+ cheap + fast + great at high volume
– can miss on hard, multi-step problems

Reasoning

prompt → thinking tokens... → answer

Good for: complex debugging, multi-step planning, hard analysis, tricky agent decisions

\+ better on hard problems
– slower – thinking tokens are billed

These are tendencies, not boxes. Many models let you set the **reasoning effort** (none → high).

rule: start with the smallest model that passes your tests. Move up only when it fails.

turning tokens into money

Price per 1M tokens

```
cost = input tokens / 1M × input price + output tokens / 1M × output price
```

EXAMPLE (illustrative prices: $0.50 in, $2.00 out)

10,000 input tokens	$0.005
1,000 output tokens	$0.002
per request	**$0.007**
× 100,000 requests / month	**$700**

REAL PRICES, OpenAI standard tier (checked 2026-10-11)

model	input / 1M	output / 1M
gpt-4o	$2.50	$10.00
gpt-4o-mini	$0.15	$0.60

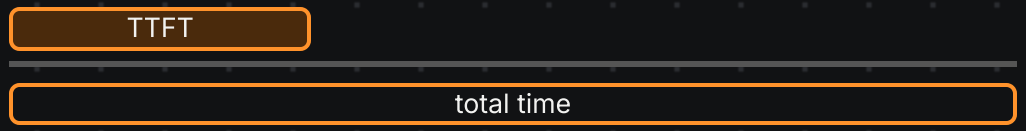
Source: developers.openai.com/api/docs/pricing

trap: compare the cost of a **correct** answer. A cheap model that needs 3 retries can cost more than a strong one.

how fast, how much

Latency and context size

Latency: two numbers



TTFT (time to first token): chat UIs
Total time: agents, APIs, batch jobs
Report **median** (typical) and **p95** (95% are faster)
Depends on: input/output length, reasoning effort, load, network, hardware

Context size: the desk

Max tokens in one request: system prompt + docs + history + question + **answer**.

`gpt-4o-mini`	128,000 context	16,384 max output

Bigger window ≠ the model uses it all well
Every extra token adds **cost and latency**
Send what's relevant (that's what RAG is for)

Source: developers.openai.com/api/docs/models/gpt-4o-mini

test on YOUR task

Why benchmarks lie

Leaderboards mislead because...

- test questions may have leaked into training data
- their task is not your task (your docs, format, language)
- models get tuned to score well on popular tests
- a 1–2 point gap is often noise
- most don't show cost or latency

So build a mini eval

1. collect 20–50 real examples with correct answers
2. run every candidate model on them
3. record accuracy, cost and latency
4. pick the cheapest model that passes your bar
5. re-run when models or prompts change

use benchmarks to build a shortlist. Use your own tests to make the decision.

where to get models

OpenRouter, Ollama, Hugging Face

OpenRouter

- A router: one key, one API, many providers' models
- OpenAI-compatible: use the OpenAI SDK
- No markup on token prices; 5.5% fee on card credit top-ups
- Free models: 50 requests/day

Ollama

- Runs open-weight models on your own machine
- OpenAI-compatible server on localhost
- Small models (1–4B) run on a laptop: llama3.2, qwen3, gemma3
- Free per token, but your hardware is the limit

Hugging Face Hub

- Where open weights are published
- Model cards: size, licence, how to run
- Download, or run through Inference Providers
- The "GitHub for models"

```
# same OpenAI SDK, only base_url changes
OpenAI()                                                    # OpenAI
OpenAI(base_url="https://openrouter.ai/api/v1", api_key=OPENROUTER_KEY)  # OpenRouter
OpenAI(base_url="http://localhost:11434/v1", api_key="ollama")           # Ollama (key ignored)
```

Sources: openrouter.ai/docs/quickstart, /docs/faq · docs.ollama.com/api/openai-compatibility · ollama.com/library · huggingface.co/docs/hub

try it yourself

Live demo: the cost of the wrong model

Route 20 support messages ("UPI failed", "GST invoice", "wrong item") into 5 categories. A **very basic** task.

```
# day-03/code
model_cost_comparison.ipynb
```

Needs OPENAI_API_KEY in day-03/code/.env

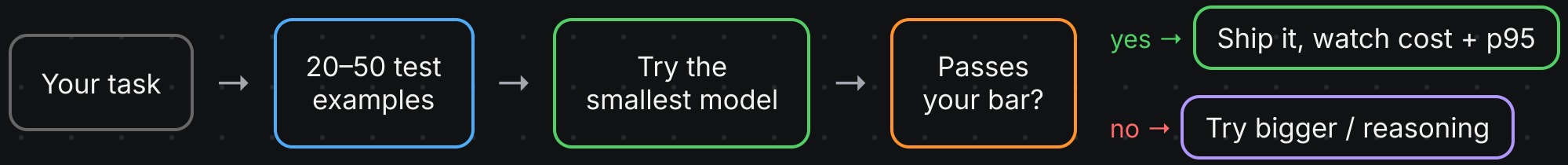
Our run (2026-10-11)

	gpt-4o	gpt-4o-mini
accuracy	100%	100%
tokens in / out	2,340 / 40	2,340 / 40
median / p95	0.94s / 1.69s	0.91s / 1.31s
1M tickets / month	**$312.50**	**$18.75**

result: same answers, **16.7× the cost**. $3,525 a year wasted at 1M tickets a month.

putting it together

How to choose a model



Ask	Look at
Can it do my task?	accuracy on **your** test set
What will it cost?	tokens per request × price per 1M × requests per month
Is it fast enough?	TTFT for chat, total time for agents, median and p95
Does my input fit?	context window and max output tokens
Am I allowed to use it?	licence, data terms, where data is processed

before next class

Homework

Comparison table: 4 models × 3 columns

Pick **one real task** from your day job (e.g. summarise a bug report, extract fields from an order email). Run it on 4 models: 1 frontier, 1 small, 1 open-weight via OpenRouter, 1 local via Ollama. Fill in:

model	quality (/10 or % correct)	speed (s)	cost per 1K requests
...			

Setup for next class: install Ollama and pull one small model; create an OpenRouter key.

ask: For your current product, would you pick a closed or an open-weight model? Why?